

Project – Phase 2: Converting ER Diagram to Relational Model

In **Phase 1**, you selected a topic, described the system, identified entities and relationships, and designed an **ER diagram**.

In **Phase 2**, you will **convert your ER diagram into a relational schema and implement it in SQL**.

Objectives

The goal of this phase is to:

- Translate ER diagrams into relational tables
- Identify **primary keys and foreign keys**
- Apply **ER-to-relational mapping rules you learned in class**
- Implement the schema using **SQL CREATE TABLE statements**

Tasks

1. Convert your ER diagrams to Tables

- Create the tables
- Define all the attributes
- Identify the **primary key** for each table

Apply the correct mapping rules depending on the relationship (one-to-one, one-to-many, many-to-one, or many-to-many) type as well as other constraints such as total participation:

Your schema should specify:

- Primary keys
- Foreign keys (when appropriate)
- NOT NULL constraints (when appropriate)

2. Write SQL Statements Needed for Creating the Tables

Write **SQL CREATE TABLE statements** for all tables in your relational model.

Your SQL should:

- Compile without errors
- Use appropriate data types
- Clearly define Primary Keys and Foreign Keys

Deliverables

Students must submit a zip file with the name format as “**Group05_phase2.zip**” where Group05 represents the group number 5. Zip file should include a report and SQL script file:

1. **Report: (File type: PDF)**
 - **Relational schema:** List of tables, Primary keys, Foreign keys
 - **Mapping explanation (short):** Explain how ER components were converted to relational tables.
2. **SQL script (File type: .sql)**
 - Create Table Statements for the entire database
3. **Appendix — AI Usage Documentation**

You are allowed (and even encouraged) to use AI tools to assist with your project. However, you must document how AI was used.

Please include the following in your report:

- Whether you used any AI tools (e.g., ChatGPT, Gemini, Claude, etc.)
- The prompts you used to help convert your ER diagram to the relational model or generate SQL code
- The AI responses you received (full responses or summarized transcripts)
- Which parts of the AI output were used directly, modified, or rejected

You may also **share a link to your AI conversation** (e.g., ChatGPT shared conversation link) instead of copying the full transcript. Make sure that the link works.